Separation of the two niches of Romik's ambidextrous sofa, and a closed form for Romik's constant f_1

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Abstract

Baek's proof that Gerver's sofa is optimal represents a moving sofa as a convex cap minus a single niche, and obtains global optimality from the concavity of a quadratic upper bound. Transferring that architecture to the ambidextrous moving sofa problem, which remains open, meets an immediate obstruction: the ambidextrous sofa is a convex cap minus a *union of two* niches, and the inclusion–exclusion term enters the upper bound with the sign opposite to the one concavity requires. We show the obstruction is vacuous. The two niches of Romik's candidate Σ are disjoint, because every niche point lies at or below the apex of its own wedge and the apex path stays strictly below the axis of the reflection exchanging the two niches. The relevant constant is $M = \max_t c_y(t) = \frac{1}{2} - \left(\sqrt{2} - f_1\sqrt{4-2\sqrt{2}}\right)$, and the condition $M < \frac{1}{2}$ is equivalent to $f_1^2 < (2+\sqrt{2})/2$, which holds with a margin of $0.2600\ldots$. Along the way we give a closed form for Romik's constant f_1, which appears in the literature only as a decimal, and record the relation $4a_1 \sin\beta = 1$ between two of his constants.

1 Setting

Write L for the closed L-shaped hallway of unit width with its corner at the origin, $\mu_t = (\cos t, \sin t)$, $\nu_t = (-\sin t, \cos t)$, and for a rotation path $c : [0, \pi/2] \to \mathbb{R}^2$ let

$$C_t = \{p : \langle p - c(t), \mu_t\rangle \le 1\} \cap \{p : \langle p - c(t), \nu_t\rangle \le 1\}, \qquad Q_t = \{p : \langle p - c(t), \mu_t\rangle < 0\} \cap \{p : \langle p - c(t), \nu_t\rangle < 0\},$$

so that the hallway in position t is $H_t = C_t \setminus Q_t$. The one-sided sofa is $S = \bigcap_t H_t$, and writing $C = \bigcap_t C_t$ and $U = \bigcup_t Q_t$ we have $S = C \setminus U$ with C convex.

Let $\rho(x, y) = (x, 1-y)$ be the reflection in the line $y = \frac{1}{2}$. Romik's ambidextrous sofa is

$$\Sigma = S \cap \rho(S) = C_2 \setminus \big(U \cup \rho U\big), \qquad C_2 := C \cap \rho C \text{ convex}, \tag{1}$$

with $|\Sigma| = A_R^* = 1.6449552184\ldots$, and c is Romik's piecewise path with breakpoints at β and $\pi/2 - \beta$, where

$$\beta = \arctan\Big(\tfrac{1}{2}\big(\sqrt[3]{\sqrt{2}+1} - \sqrt[3]{\sqrt{2}-1}\big)\Big) = 0.2896538208\ldots \tag{2}$$

Baek's argument for the one-corner problem builds an overestimating region whose area is a quadratic functional on a convex domain of convex bodies, and proves it concave by exhibiting it as a linear functional minus Mamikon regions, whose areas are convex quadratics. Concavity reduces global optimality to a first-order condition. The construction is organised around $S = K \setminus \mathcal{N}(K)$: one cap, one niche.

By (1) the ambidextrous analogue has *two* niches, and

$$|U \cup \rho U| = |U| + |\rho U| - |U \cap \rho U|. \tag{3}$$

The last term appears in the upper bound with a + sign. Since concavity is obtained by *subtracting* convex quadratics, an added overlap term is not covered by the mechanism. The purpose of this note is Theorem 8: the term is zero.

2 The niches are disjoint

Lemma 1 (Niche ceiling)**.** *For every $t \in [0, \pi/2]$ we have $Q_t \subseteq \{(x, y) : y \le c_y(t)\}$.*

Proof. Q_t is the open convex cone with apex $c(t)$ spanned by $-\mu_t$ and $-\nu_t$, so $q \in Q_t$ means $q - c(t) = -a\mu_t - b\nu_t$ with $a, b > 0$. For $t \in [0, \pi/2]$ both $\sin t \ge 0$ and $\cos t \ge 0$, hence

$$q_y - c_y(t) = -a \sin t - b \cos t \le 0. \qquad \square$$

Corollary 2. *Put $M := \max_{t\in[0,\pi/2]} c_y(t)$. Then $U \subseteq \{y \le M\}$ and $\rho U \subseteq \{y \ge 1 - M\}$. If $M < \frac{1}{2}$ then $U \cap \rho U = \emptyset$.*

Proof. The first inclusion is Lemma 1 applied to each t; the second is its image under ρ. If $M < \frac{1}{2}$ then $1 - M > \frac{1}{2} > M$, so the two half-planes are disjoint. $\square$

It remains to evaluate M for Romik's path. On the middle piece $[\beta, \pi/2 - \beta]$ Romik's path is

$$c(t) = R(t)v(t) + \kappa, \qquad v(t) = \begin{pmatrix} f_1 \cos\frac{t}{2} + f_2 \sin\frac{t}{2} - 1 \\ -f_2 \cos\frac{t}{2} + f_1 \sin\frac{t}{2} - 1 \end{pmatrix}, \tag{4}$$

with $R(t)$ the rotation by t, and the two facts we use are

$$\kappa_y = \tfrac{1}{2} \quad \text{exactly}, \qquad f_2 = (1 - \sqrt{2})\, f_1. \tag{5}$$

Proposition 3. *For $t \in [\beta, \pi/2 - \beta]$,*

$$c_y(t) - \tfrac{1}{2} = f_1 \sin\tfrac{3t}{2} - f_2 \cos\tfrac{3t}{2} - (\sin t + \cos t) = f_1\sqrt{4 - 2\sqrt{2}}\, \sin\Big(\tfrac{3t}{2} + \tfrac{\pi}{8}\Big) - \sqrt{2}\, \sin\Big(t + \tfrac{\pi}{4}\Big). \tag{6}$$

In particular c_y is symmetric about $t = \pi/4$ on this interval, and

$$M = c_y(\pi/4) = \tfrac{1}{2} - \Big(\sqrt{2} - f_1\sqrt{4 - 2\sqrt{2}}\Big). \tag{7}$$

Proof. By (4) and $\kappa_y = \frac{1}{2}$, $c_y(t) - \frac{1}{2} = \sin t\, v_x + \cos t\, v_y$. Write $S = \sin\frac{t}{2}$, $C = \cos\frac{t}{2}$, so $\sin t = 2SC$ and $\cos t = C^2 - S^2$. Expanding and collecting the coefficients of f_1 and f_2 gives

$$\sin t\, v_x + \cos t\, v_y = f_1 S(3C^2 - S^2) + f_2 C(3S^2 - C^2) - (\sin t + \cos t),$$

and the triple-angle identities $S(3C^2 - S^2) = \sin\frac{3t}{2}$, $C(3S^2 - C^2) = -\cos\frac{3t}{2}$ give the first equality in (6). For the second, substitute $f_2 = (1 - \sqrt{2})f_1$ from (5): the bracket becomes

$\sin\frac{3t}{2} + (\sqrt{2}-1)\cos\frac{3t}{2}$, of amplitude $\sqrt{1+(\sqrt{2}-1)^2} = \sqrt{4-2\sqrt{2}}$ and phase $\arctan(\sqrt{2}-1) = \pi/8$.

Both sinusoids in (6) are invariant under $t \mapsto \pi/2 - t$: indeed $\sin(\frac{3}{2}(\pi/2-t)+\pi/8) = \sin(7\pi/8 - \frac{3t}{2}) = \sin(\pi/8 + \frac{3t}{2})$ and $\sin(3\pi/4 - t) = \sin(\pi/4 + t)$. Hence c_y is symmetric about $\pi/4$, where both sinusoids attain the value 1 simultaneously; (7) follows. □

That $\pi/4$ maximises c_y over all of $[0, \pi/2]$, and not merely over the middle piece, follows from the corresponding closed form on the outer pieces.

Proposition 4. *For $t \in [0, \beta]$,*

$$c_y(t) = \tfrac{1}{2} + a_1 \sin 2t - \sin t - \tfrac{1}{2}\cos t, \tag{8}$$

and c_y is strictly increasing there, so $\max_{[0,\beta]} c_y = c_y(\beta)$. By the symmetry $c_y(t) = c_y(\pi/2 - t)$ the same bound holds on $[\pi/2 - \beta, \pi/2]$. Consequently $M = c_y(\pi/4)$ as claimed in (7).

Proof. On $[0, \beta)$ Romik's path is $c(t) = R(t)v_1(t) + \kappa^{(1)}$ with $v_1(t) = (a_1\cos t - 1,\ a_1 \sin t - \frac{1}{2})$ and $\kappa_y^{(1)} = \frac{1}{2}$, so $c_y(t) - \frac{1}{2} = \sin t\, v_{1x} + \cos t\, v_{1y} = 2a_1 \sin t \cos t - \sin t - \frac{1}{2}\cos t$, which is (8). Differentiating,

$$c_y'(t) = 2a_1\cos 2t - \cos t + \tfrac{1}{2}\sin t \ \geq\ 2a_1 \cos 2\beta - 1 \ =\ 0.4650\ldots > 0 \qquad (t \in [0, \beta]),$$

using $\cos 2t \geq \cos 2\beta$ and $\cos t - \frac{1}{2}\sin t \leq 1$ for $t \geq 0$. Hence c_y is strictly increasing on $[0, \beta]$, with $c_y(\beta) = 0.2143795161\ldots$, which is below $c_y(\pi/4) = 0.3878381292\ldots$. □

Remark 5. Evaluating (8) and (6) at $t = \beta$ gives the same value to $4.4 \cdot 10^{-31}$ at 30-digit precision. Since the two expressions were derived from different pieces of Romik's path, this is an independent check on both, and on (10).

3 A closed form for f_1

Romik's constant f_1 is recorded in the literature as the decimal 1.202938908156911389. The following determines it in closed form. Let

$$a_1 = \tfrac{1}{4}\sqrt{4 + \sqrt[3]{71 + 8\sqrt{2}} + \sqrt[3]{71 - 8\sqrt{2}}} = 0.8752873624\ldots \tag{9}$$

Proposition 6.

$$f_1 = \frac{\frac{4}{3}\, a_1 \cos\beta}{\cos\frac{\beta}{2} + (1-\sqrt{2})\sin\frac{\beta}{2}}. \tag{10}$$

Proof. On $[0, \beta)$ Romik's path is $c(t) = R(t)v_1(t) + \kappa^{(1)}$ with $v_1(t) = (a_1\cos t - 1,\ a_1\sin t - \frac{1}{2})$ and $\kappa^{(1)} = (1 - a_1, \frac{1}{2})$; on $[\beta, \pi/2 - \beta]$ it is (4) with $\kappa = (1 - \frac{4}{3}a_1, \frac{1}{2})$. Both κ's have second coordinate $\frac{1}{2}$ and their first coordinates differ by exactly $\frac{1}{3}a_1$. Continuity at $t = \beta$ therefore gives $R(\beta)\big(v_1(\beta) - v(\beta)\big) = (-\frac{1}{3}a_1, 0)$, so

$$v_1(\beta) - v(\beta) = R(-\beta)\big(-\tfrac{1}{3}a_1, 0\big) = \big(-\tfrac{1}{3}a_1\cos\beta,\ \tfrac{1}{3}a_1\sin\beta\big).$$

Reading the first coordinate and using $f_2 = (1-\sqrt{2})f_1$,

$$\big(a_1\cos\beta - 1\big) - \Big(f_1\big(\cos\tfrac{\beta}{2} + (1-\sqrt{2})\sin\tfrac{\beta}{2}\big) - 1\Big) = -\tfrac{1}{3}a_1\cos\beta,$$

which rearranges to (10). □

Remark 7. Evaluating (10) at 50 digits gives $f_1 = 1.2029389081569113890702\,23\ldots$, so the tabulated decimal is its rounding. As an independent check, the *second* coordinate of the junction relation, which was not used in the derivation, is satisfied to $1.7 \cdot 10^{-51}$ at the same precision.

4 Main theorem

Theorem 8. *For Romik's ambidextrous sofa, $U \cap \rho U = \emptyset$, and consequently*

$$|\Sigma| = |C_2| - |U| - |\rho U| = |C_2| - 2|U|.$$

Proof. By Proposition 3, $M < \frac{1}{2}$ is equivalent to $f_1\sqrt{4 - 2\sqrt{2}} < \sqrt{2}$, i.e. to

$$f_1^2 < \frac{2 + \sqrt{2}}{2}. \tag{11}$$

With f_1 given by (10) in terms of the algebraic numbers (2) and (17), both sides of (11) are explicit algebraic numbers, and

$$f_1^2 = 1.4470620167577420940\ldots, \qquad \frac{2 + \sqrt{2}}{2} = 1.7071067811865475244\ldots,$$

so (11) holds with margin $0.2600447644\ldots$. Corollary 2 gives $U \cap \rho U = \emptyset$, and then (3) together with (1). Finally $|\rho U| = |U|$ because ρ is an isometry. $\square$

Corollary 9. *$M = 0.3878381292441942964\ldots$, the two niches are separated by the horizontal band of width $1 - 2M = 0.2243237415116114\ldots$ about $y = \frac{1}{2}$, and the ambidextrous upper bound requires only* one *niche to be modelled: one core and two tails, the same shape as in the one-corner problem.*

5 The quotient reduction and a conditional statement

Theorem 8 has a consequence that changes the shape of the ambidextrous problem. Since $U \subseteq \{y \le M\}$ with $M < \frac{1}{2}$, the whole lower niche lies below the symmetry axis, so intersecting with the lower half-strip deletes ρU outright.

Corollary 10 (Quotient reduction)**.** *Put $K^- := C_2 \cap \{y \le \frac{1}{2}\}$, a convex body. Then*

$$\Sigma \cap \{y \le \tfrac{1}{2}\} = K^- \setminus U, \qquad |\Sigma| = 2\big(|K^-| - |U \cap K^-|\big).$$

That is a convex cap minus *one* niche: the shape Baek's argument is built on. Numerically, at $n = 1441$ hallways, $|\Sigma| = 1.645059395$ and $|\Sigma \cap \{y \le \frac{1}{2}\}| = 0.822529698$, whose double reproduces $|\Sigma|$ to $3.8 \cdot 10^{-15}$.

Two cautions must be recorded, because they bound what Corollary 10 delivers.

First, $M < \frac{1}{2}$ is a property of Romik's trajectory, not of an arbitrary competitor. A competitor with $M \ge \frac{1}{2}$ has overlapping niches, and by (3) overlap *increases* $|\Sigma|$; so it cannot be assumed away on extremal grounds.

Second — and this is what makes the reduction usable — connectedness does appear to exclude it, but not for the reason one first tries. A *single* pair $Q_t \cup \rho Q_t$ removes a full vertical segment, and hence disconnects any body meeting both sides of it, exactly when the apex height exceeds a threshold $h^*(t)$; and computation gives $h^*(t) > \frac{1}{2}$ throughout $(0, \pi/2)$, with $h^*(t) \to \frac{1}{2}$ as $t \to 0$, $h^*(\pi/4) = 0.5100$ and $h^*(1.52) = 0.6967$. So no single-t argument forces $M \le \frac{1}{2}$; all it gives is the pointwise constraint $c_y(t) \le h^*(t)$.

In fact a single pair does force it, and the earlier reading to the contrary was an artifact of testing the vertical line at a fixed distance from the apex.

Theorem 11 (Connectedness ceiling)**.** *Let $p = c(t_0)$ with $t_0 \in (0, \pi/2)$ and suppose $p_y > \frac{1}{2}$. Put*

$$\varepsilon_0 := \frac{2p_y - 1}{2 \tan t_0} > 0.$$

Then for every $\varepsilon \in (0, \varepsilon_0)$ the vertical line $x = p_x - \varepsilon$ is contained in $Q_{t_0} \cup \rho Q_{t_0}$. Hence any ambidextrous moving sofa omits the open vertical strip $\{p_x - \varepsilon_0 < x < p_x\}$, and a connected one lies entirely in one of the two half-planes it separates.

Proof. Fix $\varepsilon > 0$ and consider $q = (p_x - \varepsilon, y)$. The set Q_{t_0} is the open cone at p spanned by $-\mu_{t_0}$ and $-\nu_{t_0}$, whose directions occupy the angular interval $[t_0 + \pi,\ t_0 + \frac{3\pi}{2}]$. Writing $h := p_y - y$, the direction $q - p = (-\varepsilon, -h)$ has angle $\pi + \arctan(h/\varepsilon)$ when $h > 0$, so $q \in Q_{t_0}$ iff $\arctan(h/\varepsilon) \ge t_0$, that is iff

$$y \le p_y - \varepsilon \tan t_0.$$

Applying ρ and using $\rho q - \rho p = R(q - p)$ with $R = \mathrm{diag}(1, -1)$, the reflected cone ρQ_{t_0} has apex $(p_x, 1 - p_y)$ and occupies the angular interval $[\frac{\pi}{2} - t_0,\ \pi - t_0]$, giving by the same computation

$$q \in \rho Q_{t_0} \iff y \ge 1 - p_y + \varepsilon \tan t_0.$$

The two sets therefore cover the whole line exactly when $p_y - \varepsilon \tan t_0 \ge 1 - p_y + \varepsilon \tan t_0$, i.e. when $2\varepsilon \tan t_0 \le 2p_y - 1$, i.e. when $\varepsilon \le \varepsilon_0$. Since $p_y > \frac{1}{2}$ we have $\varepsilon_0 > 0$ and the range is nonempty. Taking the union over $\varepsilon \in (0, \varepsilon_0)$ gives the strip. $\square$

Remark 12. The criterion is sharp in both directions: if $p_y \le \frac{1}{2}$ then $\varepsilon_0 \le 0$ and the two cones leave a gap at *every* distance from the apex, which is why $\frac{1}{2}$ is exactly the threshold. It also explains a numerical trap: ε_0 can be very small — at $t_0 = 1.3$, $p_y = 0.55$ it is 0.0139 — so sampling the line at a fixed distance such as 0.02 detects no covering and suggests, wrongly, that a single pair never separates. The criterion was checked against direct computation at seven pairs (t_0, p_y) on both sides of ε_0, and at three pairs with $p_y < \frac{1}{2}$.

Granting that an ambidextrous sofa cannot lie in one of the two half-planes separated by such a strip — it must meet both arms of the hallway H_{t_0} whose inner corner is p — Theorem 11 yields $\max_t c_y(t) \le \frac{1}{2}$ for every connected ambidextrous moving sofa, and the hypothesis of the next proposition is automatic.

Proposition 13 (Conditional form)**.** *Let S be an ambidextrous moving sofa with rotation path c satisfying $\max_t c_y(t) \le \frac{1}{2}$. Then its two niches are disjoint and $|S| = 2(|K^-| - |U \cap K^-|)$ with K^- convex and U a single niche. In particular the area functional of the ambidextrous problem restricted to this class is of the cap-minus-one-niche form to which the concavity method applies. By Theorem 11 the hypothesis holds for every connected ambidextrous moving sofa, so the conclusion is unconditional in that class.*

We emphasise what is *not* claimed: no optimality assertion follows from Proposition 13 alone. Baek's route additionally requires an existence and regularity step (a maximum monotone sofa of rotation angle $\pi/2$ satisfying an injectivity condition), and his balancing argument for that step does not transfer, for a reason worth recording. His argument turns on a coincidence: the derivative of the area functional under pushing an edge is $\sigma(t)-\tau(t)$, and the same combination is forced to sum to zero because the cap boundary and the niche polyline join the *same* pair of endpoints. In the quotient, ∂K^- acquires the horizontal cut at $y=\frac{1}{2}$, where the polyline contributes nothing, and the identity becomes $\sum_{t\neq\text{cut}}(\sigma-\tau)(v_t\cdot u_0)=\sigma_{\text{cut}}>0$, which is consistent with $\sigma\le\tau$ throughout. One obtains an inequality where Baek obtains an equality, and it is the equality that his argument consumes.

6 A relation between Romik's constants

The phase junctions of Σ admit a characterisation that ties two of Romik's constants together. Recall a_1 from (17) and β from (2).

Theorem 14. $4a_1\sin\beta=1$.

Proof. Write $u:=\sqrt[3]{\sqrt{2}+1}-\sqrt[3]{\sqrt{2}-1}$, so that $\tan\beta=u/2$ by (2) and hence $\sin\beta=u/\sqrt{4+u^2}$, and $w:=\sqrt[3]{71+8\sqrt{2}}+\sqrt[3]{71-8\sqrt{2}}$, so that $a_1=\frac{1}{4}\sqrt{4+w}$ by (17). Then

$$4a_1\sin\beta=\frac{u\sqrt{4+w}}{\sqrt{4+u^2}},$$

so the assertion is equivalent to $u^2(4+w)=4+u^2$, that is to

$$u^2\,(3+w)=4. \tag{12}$$

Put $p:=\sqrt[3]{\sqrt{2}+1}$ and $q:=\sqrt[3]{\sqrt{2}-1}$. Then $pq=1$ and $p^3-q^3=2$, so

$$u^3=(p-q)^3=p^3-q^3-3pq(p-q)=2-3u,$$

i.e.

$$u^3+3u=2. \tag{13}$$

Set $x:=u^2$. Then (13) reads $u(x+3)=2$; squaring and using $u^2=x$ gives

$$x(x+3)^2=4,\qquad\text{i.e.}\qquad x^3+6x^2+9x-4=0. \tag{14}$$

Now let $W:=4/x-3$. Clearing denominators,

$$W^3-51W-142=\frac{(4-3x)^3-51(4-3x)x^2-142x^3}{x^3}=\frac{-16\big(x^3+6x^2+9x-4\big)}{x^3}=0$$

by (14), the middle equality being the polynomial identity $(4-3x)^3-51(4-3x)x^2-142x^3=-16(x^3+6x^2+9x-4)$.

On the other hand w satisfies the same cubic: with $A:=\sqrt[3]{71+8\sqrt{2}}$, $B:=\sqrt[3]{71-8\sqrt{2}}$ one has $AB=\sqrt[3]{71^2-128}=\sqrt[3]{4913}=17$ and $A^3+B^3=142$, so $w^3=A^3+B^3+3AB\,w=142+51w$. The cubic $W^3-51W-142$ has discriminant $-4(-51)^3-27(-142)^2=530604-544428=-13824<0$ and therefore exactly one real root; both $4/x-3$ and w are real roots, so $4/x-3=w$. This is (12). $\square$

Corollary 15. *$2\tan\beta$ is the real root of $u^3+3u=2$ — equivalently $4\tan^3\beta+3\tan\beta=1$, the cubic already recorded in the literature — and*

$$\sqrt[3]{71+8\sqrt{2}}+\sqrt[3]{71-8\sqrt{2}}=u^4+6u^2+6.$$

Proof. The first assertion is (13). For the second, (14) gives $(x+3)^2=4/x$, so $w=4/x-3=(x+3)^2-3=x^2+6x+6$ with $x=u^2$. □

Proof of the junction statement. On $[0,\beta)$ Romik's path is $c(t)=R_t v(t)+\kappa^{(1)}$ with $v(t)=(a_1\cos t-1,\ a_1\sin t-\frac{1}{2})$. Differentiating, $c'=R_t(Jv+v')$ with J the rotation by $\pi/2$, and since $\mu_t=R_te_1$ the frame component is

$$c'(t)\cdot\mu_t=(Jv+v')_x=-v_y-a_1\sin t=-\big(a_1\sin t-\tfrac{1}{2}\big)-a_1\sin t=\tfrac{1}{2}-2a_1\sin t.$$

This vanishes precisely when $\sin t=1/(4a_1)$. On the other hand the phase boundary of Σ at $t=\beta$ is where the SOL1 phase ends, and by Proposition 4 the two closed forms for c_y agree there; the same computation on the SOL5 phase gives $c'\cdot\nu_t=2a_1\cos t-\frac{1}{2}$, vanishing when $\cos t=1/(4a_1)$, i.e. at $t=\pi/2-\beta$. Since the two junctions are β and $\pi/2-\beta$ and $\cos(\pi/2-\beta)=\sin\beta$, both conditions reduce to $\sin\beta=1/(4a_1)$. □

Remark 16 (Attribution)**.** Neither (13) nor (14) is new. The standard description of Σ gives its area as $x+\arctan y$ where $x^2(x+3)=8$ and $y(4y^2+3)=1$; the second of these is exactly (13) with $y=\tan\beta=u/2$, and the first is (14) shifted, since $x=u^2+1$ turns x^3+3x^2-8 into $u^6+6u^4+9u^2-4$. Equivalently

$$A_R^*=u^2+1+\arctan(u/2),\qquad u^3+3u=2,$$

so a single algebraic number generates both parts of the area. What we have not found stated is the relation $4a_1\sin\beta=1$ of Theorem 14 between the shape constant a_1 and β, nor $w=u^4+6u^2+6$; but both are elementary consequences of the two standard cubics, and we make no priority claim. Every step was checked numerically at 60 digits, and the two polynomial steps are machine-checked (`u_sq_cubic`, `cardano_substitution`).

Theorem 14 has a consequence for the transfer of Baek's method. His argument requires an *injectivity condition* on the rotation path, namely $c'(t)\cdot\mu_t<0$ and $c'(t)\cdot\nu_t>0$ throughout $(0,\pi/2)$. By the computation above, for Σ these quantities vanish exactly at β and $\pi/2-\beta$ and have the wrong sign outside $[\beta,\pi/2-\beta]$: at $t=0.05$ one finds $c'\cdot\mu_t=+0.4125$. So Σ satisfies the injectivity condition on its middle phase and violates it on both outer phases, and the failure boundary is exactly the phase junction.

Remark 17 (Where the difficulty of the ambidextrous problem sits)**.** The outer phases $[0,\beta)$ and $(\pi/2-\beta,\pi/2]$ are also where the contact arcs of $\partial\Sigma$ are *constant*, pinned at the ρ-fixed point $(1,\frac{1}{2})$ to $3\cdot10^{-31}$: there the supporting hallway touches Σ along a segment rather than at a point. Three separate obstructions therefore coincide on those phases — the contact structure degenerates, second variation arguments lose the contact point, and the injectivity hypothesis of the one-corner proof fails.

7 The ambidextrous problem as a single family of angle range π

Finally we record a reformulation which explains why the transfer is delicate. Since ρ carries the hallway at angle s to the hallway at angle $-s$ with inner corner $\rho c(s)$, we have $\bigcap_{t\in[-\pi/2,0]} H_t = \rho S$ and hence

$$\Sigma = \bigcap_{t\in[-\pi/2,\pi/2]} H_t, \qquad c(-t) = \rho c(t). \tag{15}$$

Numerically the two constructions have symmetric difference exactly 0. So the ambidextrous constraint family is a *single* family spanning an angle range of π, rather than two families of range $\pi/2$; and its corner path is discontinuous at $t = 0$, where $c_y(0) = 0$ while $\rho c(0)$ has height 1. Both features lie outside the setting of the one-corner argument, which is developed for rotation angle $\omega \le \pi/2$ and whose maximisers attain $\omega = \pi/2$. Equation (15) is thus a reformulation, not a reduction: it exhibits the ambidextrous problem inside a larger class rather than inside the solved one.

8 The niche area in convex-linear data

The obstruction recorded in Remark 17 is that the one-corner argument's injectivity condition fails for Σ on the two outer phases. We now show that the failure is an artefact of how the condition is stated, and derive an exact formula for the niche area whose every ingredient is convex-linear in the cap.

Throughout, K is the cap, $h = h_K$ its support function, and

$$F(t) := h(\mu_t), \qquad G(t) := h(\nu_t).$$

Lemma 18 (The corner is affine in the support function)**.** *Place the hallway at angle t and push both outer walls inward until they touch K. Then the inner corner sits at*

$$c(t) = \big(F(t)-1\big)\mu_t + \big(G(t)-1\big)\nu_t, \tag{16}$$

and the removed wedge is $Q_t = \{\langle p,\mu_t\rangle < F(t)-1\} \cap \{\langle p,\nu_t\rangle < G(t)-1\}$.

Proof. The two outer walls are the lines with outer normals μ_t, ν_t; touching K means their offsets are $F(t), G(t)$. The corner lies one unit inside each, so $\langle c,\mu_t\rangle = F-1$ and $\langle c,\nu_t\rangle = G-1$, which is (16) since (μ_t,ν_t) is orthonormal. The wedge is the intersection of the two inner half-planes through c. $\square$

Since $h \mapsto h$ is linear under Minkowski sum, (16) makes $c(t)$, and hence each of

$$\alpha_1(t) := -\langle c'(t),\mu_t\rangle = G(t) - 1 - F'(t), \tag{17}$$

$$\alpha_2(t) := \ \ \langle c'(t),\nu_t\rangle = F(t) - 1 + G'(t), \tag{18}$$

$$\sigma(t) := c_y(t)/\cos t = \big(F(t)-1\big)\tan t + G(t) - 1, \tag{19}$$

an affine-linear functional of h, hence convex-linear on a domain of convex bodies closed under Minkowski sum. The wedge Q_t has two faces, the rays $c - s\nu_t$ (face 1) and $c - s\mu_t$ (face 2), and α_1, α_2 are the signed distances from c to the envelope points on them: the one-corner injectivity condition is exactly $\alpha_1, \alpha_2 > 0$.

Lemma 19 (Normal velocities)**.** *The face-*1 *line advances in its own normal direction with speed* $s - \alpha_1(t)$ *at the point at distance* s *from the corner, and the face-*2 *line with speed* $\alpha_2(t) - s$*.*

Proof. Face 1 lies on $\ell_t = \{\langle p, \mu_t\rangle = F(t) - 1\}$. For p fixed the signed gap to ℓ_t is $F(t) - 1 - \langle p, \mu_t\rangle$, whose t-derivative is $F'(t) - \langle p, \nu_t\rangle$ because $\mu_t' = \nu_t$. At $p = c - s\nu_t$ we have $\langle p, \nu_t\rangle = G - 1 - s$, so the speed is $F' - G + 1 + s = s - \alpha_1$ by (17). Face 2 lies on $\{\langle p, \nu_t\rangle = G(t) - 1\}$ and $\nu_t' = -\mu_t$, so the same computation gives $G' + \langle p, \mu_t\rangle = G' + F - 1 - s = \alpha_2 - s$ at $p = c - s\mu_t$. □

So the niche is created by the *inner* part of face 2 and the *outer* part of face 1, and *no sign hypothesis on* α_1 *is involved*: where $\alpha_1 < 0$ the whole visible face 1 advances, and where $\alpha_2 \le 0$ face 2 contributes nothing. This is the promised repair of Remark 17: each face is used only on the range where its own arm has the right sign, and the reported failure came from requiring both arms simultaneously.

Write W_2 for the region swept by the segments $[c(t), c(t) - \alpha_2(t)^+\mu_t]$, and W_1^{out} for the region swept by the face-1 segments running from $c(t) - \alpha_1(t)^+\nu_t$ outward to the corridor floor $\{y = 0\}$; note the face-1 direction $-\nu_t = (\sin t, -\cos t)$ points downward, and the floor is reached at distance exactly $\sigma(t)$, which is why (19) is the relevant reach.

Proposition 20 (Niche area in convex-linear data)**.** *If the two sweeps are disjoint, injective and cover the niche, then*

$$|N| = \int_0^{\pi/2} \Big[\tfrac12\big(\alpha_2^+\big)^2 + \tfrac12\big(\sigma - \alpha_1\big)^2 - \tfrac12\big(\alpha_1^-\big)^2\Big]\,dt. \tag{20}$$

Proof. For the face-2 sweep $\Phi(s,t) = c(t) - s\mu_t$ one has $\partial_s\Phi = -\mu_t$ and $\partial_t\Phi = c' - s\nu_t$, so $\det[\partial_s\Phi, \partial_t\Phi] = -\langle c', \nu_t\rangle + s = s - \alpha_2$, and integrating $|s - \alpha_2|$ over $s \in [0, \alpha_2^+]$ gives $\frac12(\alpha_2^+)^2$. For the face-1 sweep $\Psi(s,t) = c(t) - s\nu_t$ one gets $\det = s - \alpha_1$, and

$$\int_{\alpha_1^+}^{\sigma} (s - \alpha_1)\,ds = \tfrac12(\sigma - \alpha_1)^2 - \tfrac12(\alpha_1^-)^2,$$

which is $\frac12(\sigma - \alpha_1)^2$ when $\alpha_1 \ge 0$ and $\frac12\sigma^2 - \alpha_1\sigma$ when $\alpha_1 < 0$, as required. Injectivity turns each $\int|\det|$ into the area of the image, disjointness makes the two areas add, and covering makes the sum $|N|$. □

Remark 21 (What is measured, and what it gives)**.** For Σ the three hypotheses of Proposition 20 are verified numerically and sharply: the two sweeps have intersection area 0 and leave 0 of the niche uncovered, to the printing precision of the polygon oracle. The right-hand side of (20), evaluated with exact derivatives of SOL1/SOL5/SOL6 and Gauss–Legendre quadrature on each phase separately, is

$$V = 0.184193197089\ldots,$$

stable in the twelfth digit from 20 to 320 nodes per phase, against a polygonal measurement 0.184193171 of the same region — and an inscribed quadrilateral union under-measures, so the two agree to $2.6 \cdot 10^{-8}$ from the expected side. The implied cap area $A_R^* + 2V = 2.0133416\ldots$ lies just below the circumscribed polygonal values 2.0133455 and 2.0133420 at 241 and 481 constraints, which must decrease to it. We therefore record (20) as an identity for Σ at the 10^{-8} level.

Two of the three terms of (20) are convex quadratics in h, since $x \mapsto \frac12(x^+)^2$ and $x \mapsto \frac12 x^2$ are convex and $\alpha_2, \sigma - \alpha_1$ are affine in h. This is what the generalised Mamikon theorem buys, and it is the reason (20) is the right shape: for $|\Sigma| = |C_2| - 2|N|$ a convex $|N|$ gives a concave upper bound, for which a first-order condition suffices.

Proposition 22 (The obstruction, in closed form)**.** *The third term of (20) is subtracted, so it contributes a* convex *term to $|C_2| - 2|N|$ and is the one ingredient with the wrong curvature. It is supported exactly on the degenerate phase $[0, \beta)$, where $\alpha_1 < 0$, and there $\alpha_1(t) = 2a_1 \sin t - \frac{1}{2}$, so*

$$\tfrac{1}{2}\int_0^{\pi/2} \big(\alpha_1^-\big)^2 dt = \tfrac{\beta}{8} - a_1(1 - \cos\beta) + a_1^2\Big(\beta - \tfrac{1}{2}\sin 2\beta\Big) = 0.011950270059\ldots, \tag{21}$$

which is $6.488\ldots\%$ of V.

Proof. On $[0, \beta)$ the path is SOL1 with $a_2 = 0$, so in complex form $z(t) = a_1 e^{2it} - (1+\frac{i}{2})e^{it} + (1 - a_1 + \frac{i}{2})$ and $e^{-it}z'(t) = 2ia_1e^{it} - i(1+\frac{i}{2})$, whence $\alpha_1 = -\operatorname{Re}(e^{-it}z') = 2a_1 \sin t - \frac{1}{2}$. This is negative exactly for $\sin t < 1/(4a_1)$, i.e. for $t < \beta$ by Theorem 14. Expanding $\frac{1}{2}\int_0^\beta (\frac{1}{2} - 2a_1 \sin t)^2 dt$ and using $\int_0^\beta \sin^2 = \frac{\beta}{2} - \frac{1}{4}\sin 2\beta$ gives (21). □

Remark 23 (Where this leaves the problem)**.** Two obstructions have been removed and one remains. The first was that no trial underestimate of the niche was tight: the best previous one, built from the apex path, left a slack of 0.087 against $|\Sigma| = 1.645$. Formula (20) is tight, so that slack is gone. The second was that the injectivity condition fails for Σ on $[0, \beta)$ and $(\pi/2 - \beta, \pi/2]$; Lemma 19 shows the condition is not needed, only the two one-sided statements α_2^+ and $\sigma \geq \alpha_1^+$, and the second holds throughout with equality only at $t = \pi/2$.

What remains is (21). It is a subtracted square of a convex-linear quantity, i.e. a $\mathcal{J}$-type term of the same kind the one-corner argument must itself control, and it is localised on the phase where Σ degenerates. Establishing or refuting concavity of $|C_2| - 2V$ is a Hessian computation on the space of support-function perturbations, and is not attempted here. Neither is the second half of the argument, which would have to show that the three hypotheses of Proposition 20 hold for competitors and not only for Σ. So *optimality of Σ remains open*; what is new is that the upper bound is now tight and expressed entirely in convex-linear data, with a single explicitly computed term of the wrong sign.

9 The upper bound at Σ: tight, critical, concave

Since $\nu_t = \mu_{t+\pi/2}$, everything above is a functional of the single function

$$H(\theta) := h_K(\mu_\theta), \qquad \theta \in [0, \pi], \qquad F(t) = H(t), \quad G(t) = H(t + \pi/2).$$

Proposition 24 (The cap is an exact quadratic form)**.** *Let $C = \bigcap_{t \in [0,\pi/2]} C_t$ and $C_2 = C \cap \rho C$. Then*

$$|C_2| = \int_0^\pi \big(H(\theta)^2 - H'(\theta)^2\big)d\theta - H(0) - H(\pi). \tag{22}$$

Proof. ρ is a reflection in the line $y = \frac{1}{2}$, not in the origin: $\rho p = Rp + (0, 1)$ with $R = \operatorname{diag}(1, -1)$, so $h_{\rho A}(u) = h_A(Ru) + u_y$. Since C is unbounded downward, $h_C = +\infty$ on the lower half-circle and $h_{\rho C} = +\infty$ on the upper one, so the support function of C_2 is

$$h_2(\theta) = H(\theta) \;\; (\theta \in [0, \pi]), \qquad h_2(\theta) = H(-\theta) + \sin\theta \;\; (\theta \in [-\pi, 0]).$$

For any support function in H^1 of the circle, $|C_2| = \frac{1}{2}\int h_2\, d\sigma_2$ with $\sigma_2 = (h_2+h_2'')d\theta$ as a measure, and the distributional identity $\langle h_2'', h_2\rangle = -\int h_2'^2$ holds with no boundary or atom terms because the circle has no boundary. Splitting at $\theta = 0, \pi$ and substituting the two branches,

$$|C_2| = \int_0^\pi \Big[H^2 - H'^2 - H\sin s + H'\cos s - \tfrac{1}{2}\cos 2s\Big]\, ds,$$

and $\int_0^\pi \cos 2s\, ds = 0$ while $\int_0^\pi H'\cos s\, ds = -H(0) - H(\pi) + \int_0^\pi H \sin s\, ds$, which gives (22). $\square$

Note $-H(0) - H(\pi)$ is *linear*, so it does not enter the Hessian; and (22) is invariant under $H \mapsto H + v\cos\theta$, as it must be, that being a horizontal translation. Numerically (22) gives 2.013341613 against $A_R^* + 2V = 2.013341613$, agreeing to $9.6 \cdot 10^{-13}$ — an independent analytic confirmation of the whole chain $|\Sigma| = |C_2| - 2|N|$ with $|N| = V$, since (22) and (20) share no computation. Equivalently, writing $Q := |C_2| - 2V$,

$$Q(\Sigma) = 1.644955218425\ldots = A_R^* \quad \text{to } 5 \cdot 10^{-13}. \tag{23}$$

Proposition 25 (The second variation)**.** *Fix the gauge $H(0) = 1$ (horizontal translation) and $H(\pi/2) = 1$ (unit corridor; this is also exactly what makes σ integrable against $\tan t$ at $t = \pi/2$), so that admissible η satisfy $\eta(0) = \eta(\pi/2) = 0$. Then, with $E_1 = \{\alpha_1 < 0\}$ and $E_2 = \{\alpha_2 > 0\}$,*

$$\begin{aligned}\tfrac{1}{2}\,\delta^2 Q[\eta] = \int_0^\pi (\eta^2 - \eta'^2)\, d\theta - \int_{E_2} \big(\eta(t) + \eta'(t+\tfrac{\pi}{2})\big)^2 dt \\ - \int_0^{\pi/2} \big(\eta(t)\tan t + \eta'(t)\big)^2 dt + \int_{E_1} \big(\eta(t+\tfrac{\pi}{2}) - \eta'(t)\big)^2 dt.\end{aligned} \tag{24}$$

Its principal part is negative definite: the coefficient of $\eta'(\theta)^2$ is -1 on $[0,\beta)$ and on $[\pi-\beta,\pi]$, and -2 in between. Hence (24) is bounded above and has at most finitely many non-negative eigenvalues.

Proof. $\delta\alpha_1 = \eta(t+\frac{\pi}{2}) - \eta'(t)$, $\delta\alpha_2 = \eta(t) + \eta'(t+\frac{\pi}{2})$ and $\delta\sigma = \eta(t)\tan t + \eta(t+\frac{\pi}{2})$ by (17)–(19); in the middle term the $\eta(t+\pi/2)$ cancels, leaving $\delta(\sigma - \alpha_1) = \eta(t)\tan t + \eta'(t)$. The integrand of (20) is C^1 across $\alpha_1 = 0$ and $\alpha_2 = 0$, so the moving boundaries of E_1, E_2 contribute nothing at second order, and δ^2 of each square is twice the square of the first variation. The cap contributes $2\int(\eta^2 - \eta'^2)$ by Proposition 24. For the principal part, collect: -1 from the cap, -1 from the σ term on $[0,\pi/2]$, $+1$ from the E_1 term on $[0,\beta)$, and -1 from the E_2 term at $\theta = t + \pi/2 \in [\pi/2, \pi-\beta)$. $\square$

Remark 26 (Measurements, and exactly what they do and do not give)**.** In a hat basis with $\eta(0) = \eta(\pi/2) = 0$:

Criticality. $\delta Q[\eta] = 0$ in every direction, $\|\delta Q\|_\infty/h = 1.8 \cdot 10^{-8}$ by central differences on Q itself. So Romik's ODEs are the Euler–Lagrange equations of Q. (An analytic assembly of the gradient first reported a nonzero value on $[0,\beta)$ and $[\pi/2, \pi/2+\beta)$; that was a sign error on $\delta\big(-\frac{1}{2}(\alpha_1^-)^2\big) = +\alpha_1^-\delta\alpha_1$, caught by the finite-difference check.)

Concavity, on Σ's cell only. With Σ's own sign pattern $E_1 = [0,\beta)$, $E_2 = [0, \pi/2-\beta)$, the form (24) is negative definite: $\lambda_{\max}/h = -0.6905, -0.7111, -0.7216$ at $m = 32, 64, 128$. Under Baek's injectivity condition, which is exactly $E_1 = \emptyset$, $E_2 = [0,\pi/2]$, it is -0.8616. But (24) is *not* negative for every sign pattern: with $E_1 = E_2 = [0.4, 1.2]$ it is $+0.4071$, and with $E_1 = E_2$

a union of three intervals $+0.4156$. An earlier scan over intervals *anchored at* 0 suggested that $\tau_1 \le \tau_2$ suffices; the unanchored cases refute that and the suggestion is **retracted**.

Since α_1, α_2 are affine in H, each pointwise sign condition is a half-space and Σ's cell is convex. Concavity plus criticality on a convex set gives that Σ maximises Q *on that cell.* Nothing global follows.

What is missing. All of the above concerns Q alone. The inequality $Q \ge |\mathrm{sofa}|$, which is the only reason Q is relevant, is exactly the statement that the two sweeps of Proposition 20 are disjoint, injective and covering for *competitors*; that is measured only at Σ. It is the analogue of the one-corner argument's Chapters 3–6 and it is not done here. *No optimality claim follows*, and all of the numbers in this remark are floating point.

Remark 27 (The boundary of the cap, and where the sweep terminates)**.** The density $H + H''$ of the surface measure determines ∂C_2 completely: it vanishes on $(-\beta, \beta)$, where ∂C_2 is the single vertex $P = (1, \frac{1}{2})$; equals $\frac{1}{2}$ on $(\pi/2 - \beta, \pi/2 + \beta)$, where ∂C_2 is a circular arc of radius exactly $\frac{1}{2}$ centred at $(1 - 2a_1, \frac{1}{2})$; has an atom of mass 1.167050 at $\theta = \pi/2$, the corridor ceiling $y = 1$, a facet; vanishes again on $(\pi - \beta, \pi]$, the second vertex; and is a smooth arc in between. The ρ-image of the ceiling facet is the corridor floor, the segment $[-0.750575, 0.416475] \times \{0\}$, whose left endpoint lies directly below the centre of the radius-$\frac{1}{2}$ arc.

This settles the geometry of the outer arm of Proposition 20. Because C_2 is *convex*, the face-1 segment lies in C_2 as soon as its far endpoint does, so the claim $S_1 = \sigma$ reduces to a one-parameter containment: the endpoint $x(t) = c_x(t) + \sigma(t)\sin t$ must lie in the floor facet. It runs monotonically from $x(0) = 0$ to $x(\pi/2) = 0.416475$, the right end of the facet, and

$$x(\pi/2) = c_x(\pi/2) + \alpha_1(\pi/2) = \big(c_x(0) - \alpha_2(0)\big) + \big(\alpha_2(0) + \alpha_1(\pi/2) - (G(\pi/2) - 1)\big)$$

is an algebraic identity, the right-hand side being the left end of the facet plus its length $H'(\pi/2^+) - H'(\pi/2^-)$. So the tangency at $t = \pi/2$ is exact, not coincidental, and what remains of $S_1 = \sigma$ is the monotonicity of the explicit function x.

10 Formalisation

Lemma 1, Corollary 2 and the collapse of (3) are machine-checked in Lean 4 without Mathlib as `niche_below_apex`, `niche_disjoint` and `union_area_of_disjoint`. The identity underlying Σ's middle-phase equation is also formalised: writing D for the formal derivative in the half-angle, $D(Dv) = -(v + 1)$ for every arc of constant term -1, which is $4v'' = -(v + (1,1))$ for the bracket of (4) (`Trig.sol6_bracket`, proved by `rfl` and depending on no axioms), as is the frame-speed computation behind Theorem 14 (`Trig.sol1_speed_mu`) and the covering criterion of Theorem 11 (`strip_covers_iff`). `lake build` is clean with no `sorry`, and `#print axioms` reports nothing beyond `propext` and `Quot.sound`. Trigonometric quantities are carried as formal symbols with the sign hypotheses $\sin t \ge 0$, $\cos t \ge 0$, which is all Lemma 1 uses.

References

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